Federated Distillation in Massive MIMO Networks: Dynamic Training, Convergence Analysis, and Communication Channel-Aware Learning

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***Abstract*—Federated Distillation (FD) is a novel distributed learning paradigm that shares the privacy-preserving nature of Federated Learning (FL) and provides possible solutions to the challenges introduced by the FL framework, such as being able to train local models with nonidentical architectures. In this paper, a communication channel-aware FD-framework is presented for multi-user massive multi-input-multi-output (mMIMO) communication system, where zero-forcing (ZF) and minimum mean-squared-error (MMSE) schemes are utilized to null the intra-cell interference. Unlike most existing studies, where both model parameters and model outputs (logits) are utilized for transmission, we exclusively adopt logits as the information exchanged in the wireless links to reduce the overall communication overhead in each round. Based on the analysis, the dynamic training steps based FD algorithm (FedTSKD) is proposed to save communication resources and accelerate the training process. Further, a group-based FD algorithm (FedTSKD-G) is proposed for the system experiencing different channel conditions like deep-fade. Simulation results on image classification tasks with ImageNette/STL-10, CIFAR-10/STL-10 and MNIST/FMNIST datasets combinations have demonstrated the proposed algorithm's effectiveness and efficiency. Comparison with the FL algorithm shows that the proposed FD algorithm only incurs 1% of FL's communication overhead to achieve the same testing performance.**

***Index Terms*—Deep learning, federated distillation, federated learning, massive MIMO (mMIMO).**

I. Introduction

BEING recognized as a promising distributed learning algorithm, Federated Learning (FL) [1] is able to decouple the processes of data acquisition and computation. Such privacy-preserving nature has been investigated and analyzed in existing literatures [2], [3], [4], [5]. One significant research direction in this topic is integrating FL frameworks with appropriate communication-efficient systems and fully empowering the potentials provided by next-generation communication systems. Meanwhile, all the clients participating in the FL training process must retain local models with the same architecture [6]; FL lacks model heterogeneity among local users.

Recently, by leveraging the knowledge distillation (KD) technique [7], Federated Distillation (FD) [8] has been attracting ever-increasing attention from both industry and academia. Unlike FL, FD allows local users to train nonidentical local models. Instead of model parameters, output logits on the public dataset are utilized as the information exchanged between BS and local users. This not only grants local users more flexibility to trade off their local computing capability and training performance but also reduces the incurred communication cost remarkably; achieving both heterogeneity and efficiency.

In the wake of the swift deployment of fifth-generation (5G) communication infrastructures and networks at a global scale, both industrial and academic sectors have shifted their research focus towards the exploration of the next-generation air interface (NGAI), including beyond 5G and sixth-generation (6G) communications. In contrast to the paradigm established by 5G, the landscape of 6G communications necessitates the comprehensive integration of Artificial Intelligence (AI) throughout heterogeneous and expansive networks, aspiring to realize the concept of ubiquitous AI within the 6G domain [9]. Recognized for its privacy-preserving attributes, FD emerges as a particularly promising solution poised to address the inherent challenges in 6G communication networks [10]. Meanwhile, the advantages provided by the NGAI, such as high-performance networking, hyper-flexible architecture and AI-enhanced security [11], are also vital for accomplishing the requirements of FD frameworks in future communication networks. In particular, the advantages inherent in NGAI massive MIMO systems, such as spatial multiplexing, diversity [12], interference management [13], and energy efficiency [14], are poised to reciprocally benefit the FD algorithm, especially in resource-constrained and heterogeneous environments.

A. Knowledge Distillation-Assisted FL/FD Frameworks

Some of the existing works have endeavored to incorporate the KD technique in the vanilla FL algorithm and improve corresponding system performance by introducing new flexible aggregation schemes [15], [16], [17] or proposing enhanced

Manuscript received 20 September 2023; revised 9 January 2024; accepted 10 March 2024. Date of publication 18 March 2024; date of current version 8 August 2024. This work was supported by the UK Engineering and Physical Sciences Research Council (EPSRC) under grant number EP/T021063/1. The associate editor coordinating the review of this article and approving it for publication was J. Zhao. *(Corresponding author: Yuchen Mu.)*

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Digital Object Identifier 10.1109/TCCN.2024.3378215

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TABLE I
COMPARISON WITH FD SYSTEMS. ABBREVIATIONS IN THE TABLE ARE DEFINED AS: "C": CONVERGENCE ANALYSIS, "MH": MODEL HETEROGENEITY, $\mathcal{X}$ IS THE UNLABELED PUBLIC DATASET THAT BOTH LOCAL USERS AND SERVER HAVE ACCESS TO, "SD": SERVE SIDE DISTILLATION, "LD": LOCAL USER SIDE DISTILLATION, "IE": INFORMATION EXCHANGED, $\mathcal{W}$ DENOTES THAT MODEL WEIGHTS ARE TRANSMITTED, $\mathcal{L}$ DENOTES THAT MODEL OUTPUT LOGITS ARE TRANSMITTED, "N": RECEIVER SIDE NOISE, "F": CHANNEL FADING, "I": INTERFERENCE FROM OTHER USERS

Refs.	C/MH/$\mathcal{X}$	SD/LD	IE-UL	IE-DL	UL-N/F/I	DL-N/F/I
[15]	× ✓ ✓	✓ ✓	$\mathcal{W}$	$\mathcal{W}$	× × ×	× × ×
[16]	× ✓ ✓	✓ ×	$\mathcal{W}$	$\mathcal{W}$	× × ×	× × ×
[17]	× ✓ ×	× ✓	$\mathcal{L}$	$\mathcal{L}$	× × ×	× × ×
[18]	× ✓ ✓	× ✓	$\mathcal{L}$	$\mathcal{L}$	× × ×	× × ×
[19]	✓ × ✓	✓ ✓	$\mathcal{W}$	$\mathcal{W}$	× × ×	× × ×
[20]	✓ ✓ ×	× ✓	$\mathcal{W}$	$\mathcal{W}$	× × ×	× × ×
[21]	× ✓ ×	✓ ✓	$\mathcal{L}$	$\mathcal{W}$	× × ×	× × ×
[22]	× ✓ ×	× ✓	$\mathcal{W}$	$\mathcal{W}$	× × ×	× × ×
[23]	× × ✓	✓ ×	$\mathcal{L}$	$\mathcal{W}$	✓ ✓ ×	✓ ✓ ×
[24]	× × ×	× ✓	$\mathcal{L}$	$\mathcal{L}$	✓ ✓ ×	× × ×
[25]	× × ×	× ✓	$\mathcal{L}$	$\mathcal{L}$	✓ ✓ ×	✓ ✓ ×
[26]	× × ✓	✓ ×	$\mathcal{L}$	$\mathcal{W}$	× × ×	× × ×
[27]	× × ✓	× ✓	$\mathcal{L}$	$\mathcal{L}$	× × ×	× × ×
[28]	× ✓ ✓	✓ ✓	$\mathcal{L}$	$\mathcal{L}$	✓ × ×	✓ × ×
[29]	× × ✓	× ✓	$\mathcal{W}$	$\mathcal{W}$	✓ × ×	✓ × ×
Prop.	✓ ✓ ✓	✓ ✓	$\mathcal{L}$	$\mathcal{L}$	✓ ✓ ✓	✓ ✓ ✓

local distillation/training algorithm [18], [19], [20], [21], [22]. Specifically, work in [15] proposes a weighted ensemble distillation algorithm based on private certainty scoring. In [16], in order to mitigate privacy risk, a flexible aggregation scheme is designed and unlabeled datasets are utilized for model fusion. Likewise, the aggregation methodology introduced in [17] leverages the statistical characteristics of uploaded logits and stays robust to the local model corruption issue, where the slow convergence of the local model belonging to one individual user might disseminate its effects to other local users. On the other hand, the work in [18] exploits the transfer learning technique and distillation was only performed at local user side. In [19], a generative model is learnt at server side and broadcasted to local users such that user information is ensembled and aggregated in a data-free manner. In [20], a divergence-based loss function is proposed for the distillation process, and the corresponding convergence analysis is also provided. Similarly, a feature distillation loss is introduced in [21] employing new trainable parameters, where classification and feature-similarity losses are combined and optimized jointly. The work in [22] designs a new variant of vanilla FD algorithm where each user is required to maintain two local models; that is, teacher model and student model, and only the student models will participate in the training process collaboratively and be shared.

In these works, practical communication aspects are completely ignored (also see in Table I), which typically include quantization, modulation and transmission. At the same time, model parameters are still the information to be exchanged between BS and local users; that is, the communication-efficiency offered by FD paradigm is not exploited in these works (the works in [17] and [18] utilize the logits as the exchanged information for both UL and DL-links, but both of the proposed algorithms require a pre-training phase at local user side, which will also introduce extra communication overhead and latency).

B. Communication-Integrated FD Frameworks

Meanwhile, in the direction to integrate vanilla FD paradigm with communication-efficient systems, existing studies are summarized as follows. In [23], in order to establish communication-efficiency promotion, uplink-downlink capacity asymmetry issue is investigated, where model parameters and model outputs are considered as the exchanged information for DL and UL transmission, respectively. Besides, a mixup technique is employed before local users upload the latest model outputs to further preserve privacy, and model distillation is only performed at server side. Similarly, the work in [24] considers both digital and analog transmission schemes, where Gaussian multiple-access channels are utilized as UL communication link model, leaving DL to be error-free. At the same time, [24] assumes no global model at server side; that is, distillation is only performed at local user side and the BS is only responsible for carrying out logits aggregation. The work in [24] was extended by the same authors later in [25] via considering imperfect DL channels. The works in [26], [27], [28], [29] manage to improve communication efficiency by reducing the total UL transmission overhead. Specifically, the works in [26], [27] endeavor to design new schemes such that only the logits with low entropy will be selected, transmitted and participate in the aggregation procedure within each global communication round. Similarly, with same design idea, a soft-label quantization and source coding schemes are proposed and implemented in [28] to reduce the entropy of logits. In [29], a dynamic quantization scheme, where appropriate quantization levels are utilized by different users, is proposed and utilized to carry out model compression and reduce overall communication cost for the UL links.

In Table I, we summarize and compare the algorithms mentioned in this paper, the comparison is made for convergence analysis, model heterogeneity, information exchanged via channels, location that distillation is performed, categories of noises considered in the communication process, etc. Specifically, for the communication-integrated frameworks discussed in this subsection [23], [24], [25], [26], [27], [28], [29], it can be witnessed that none of them are designed and investigated under a systematic communication scenario with the presence of all types of noises. To name a few, the works in [23] and [25] only consider channel fading and receiver side noise, leaving interference from other users uninvestigated. In the same way, the work in [24] only includes the receiver side noise and channel fading in the UL transmission, leaving DL to be error-free. Besides, the works in [23], [24], [25], [26], [27], [29] assume all the users to have a local model with identical structure; that is, the model heterogeneity offered by the FD algorithm is not exploited. On the other hand, even though the work in [28] empowers the local users to maintain models with different architectures, it only takes the imperfections introduced by quantization noise into consideration and corresponding convergence analysis is not conducted. Further, although the works in [19] and [20]

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provide corresponding convergence analysis, model weights are still treated as the exchanged information between local users and server; that is, the convergence bounds derived are for FL algorithm, rather than FD algorithm.

C. Contributions

In this paper, we propose a communication-efficient FD algorithm under a practical multi-user massive multi-input-multi-output (mMIMO) communication system. To provide communication efficiency, the proposed algorithm focuses on regulating the power for uplink and downlink, intelligently changing the number of training steps per communication round, and varying the quantization and modulation. Unlike the existing works listed in [15], [16], [17], [18], [19], [20], [21], [22], where communication channels are assumed perfect and interference-free, we consider both UL and DL communication links to be error-prone while considering modulation and quantization. To the best of authors' knowledge, we are the first to evaluate FD algorithm under a mMIMO communication system and prove the corresponding convergence. The main contributions of this paper can be summarized as follows.

1) Massive MIMO-Integrated FD Algorithm (FedTSKD): We design and implement a FD algorithm with massive MIMO communication network, accounting imperfections in both UL and DL links. To mitigate the multi-user interference, we utilize zero-forcing (ZF) and minimum mean-squared-error (MMSE) approaches for both uplink and downlink. In the proposed work, logits are treated as the only information exchanged over the imperfect communication links.

We carry out extensive simulations to demonstrate the effects of various parameters on the FD convergence, testing accuracy, and communication efficiency; these parameters include number of antennas, number of users, UL and DL SNRs, channel fading, quantization, modulation, noise, intra-cell interference, different degrees of non-i.i.d.-ness of available datasets.

Further, we compare simulation results of the proposed algorithm with two baselines in the existing literature: *1)* Both communication links (DL and UL) are assumed to be error-free [15], [16], [17], [18], [19], [20], [21], [22]; *2)* Only uplinks are error-prone, and downlinks error-free [24]. Moreover, we also compare two popular learning algorithms, viz, FL and FD algorithms, where both are integrated with massive MIMO system. Simulations show the FD algorithm incurs 1% of the communication overhead of the FL algorithm, which is significant.

2) Convergence Analysis for the Proposed FD Algorithm: We provide the convergence analysis for the proposed communication efficient FD algorithm, accounting for imperfect links. The proposed analysis shows that the modeling error of the FD algorithm reduces in $\mathcal{O}(T^{-1})$ manner, given the parameters of the learning algorithm satisfy certain conditions. We also show that given the conditions on the learning rate, the number of distillation steps can be adjusted to obtain better convergence bound of the modeling error.

3) Group-Based FD Algorithm (FedTSKD-G): Motivated by the heterogeneity of wireless channels, we propose a group-split scheme, which allows the server side model to be aware of the effects resulting from different channel imperfections while making predictions on the public dataset. In addition to that, by transferring the design ideas on adapting the number of local steps, we reduce the overall communication cost incurred during the training process and accelerate the convergence. Simulation results on image classification problem with ImageNette/STL-10, CIFAR-10/STL-10 and MNIST/FMNIST dataset combinations demonstrate both the effectiveness of the proposed framework.

Remark (Difference to our recent work [30]): The work in [30] is based on Federated Learning, whereas the present work focuses on Federated Distillation. We summarize the differences between these two works as follows.

- *Mathematical analysis perspective:* Two different algorithms (FD and FL) have different convergence analyses, which are based on logits and weights respectively. Compared with FL setting, the convergence analysis for FD paradigm in mMIMO system is much more challenging. In both cases, wireless impairments such as fading, interference and noise introduced by the communication transmission process are considered. Intuitively, in FL scenario, the perturbations in model weights will have a discernible impact on the overall model performance. While in the FD setting, the wireless impairments are imposed on the model output (logits), as will be presented in Section IV, the underlying relationship between the logits and system performance is more complicated to derive, and this is because, unlike the FL algorithm, in FD, there is no "*direct*" connection between these two factors and this calls for joint consideration of both training and distillation processes.
- *Algorithm design perspective:* The design focus of our previous work [30] is about improving the *communication efficiency*. In this work, apart from this, we also emphasize *communication-aware* mechanism. Specifically, let the system be conscious of the communication quality and learn the imperfections introduced by the wireless channels itself. This will be elaborated in detail in Section V.
- *Simulation perspective:* For the simulations in [30], all local users' model architecture has to be the same, this neglects the heterogeneity in terms of local users' computing capability. In this work, different clients' models are considered to be nonidentical to fully exploit the heterogeneity provided by the FD paradigm. Further, more challenging datasets, have been considered in this work to ensure and verify the proposed FD algorithm and its convergence.

Organization: The rest of this paper is organized as follows. Sections II and III detail the proposed system model and communication process included in the proposed framework, respectively. Convergence analysis is conducted in Section IV, followed by the explanation of the proposed communication channel-aware FD scheme in Section V. Simulation results are presented in Section VI, and finally Section VII concludes this paper.

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TABLE II
SUMMARY OF THE NOTATIONS USED IN THIS PAPER

Symbols	Notations
n, N	User index and total number of users
$\mathbf{x}, \mathbf{y}$	Input and output of a data sample
$\mathcal{D}_n, \mathcal{D}$	Local private dataset for user n and all users
D_n, D	Size of local private dataset for user n and all users
$\mathcal{X}, D_p$	Unlabeled public dataset and its size
$\mathbf{w}, \mathbf{w}^*$	Arbitrary and optimal model parameters
$\mathbf{z}$	Output logits
F_n	Mapping between input samples and output logits
$l(\cdot)$	Training loss function
$L_n(\cdot)$	Overall training loss function for user n
$l_{KL}(\cdot)$	Kullback-Leibler divergence loss
$Q_n(\cdot)$	Overall distillation loss function for user n
ψ	Smooth function factor for L_n
μ, μ_q	Strongly convex function factors for L_n and Q_n
$\sigma_n^2, \sigma_{qn}^2$	Factors of bounded SGD variance for L_n and Q_n
G^2, G_q^2	Factors of bounded gradients for L_n and Q_n
t, T	Time step index and total number of time steps
$\mathcal{T}$	Set for all time steps
$\mathcal{T}_{LT}$	Sets of time indices for local training
$\mathcal{T}_{SD}, \mathcal{T}_{LD}$	Sets of time indices for server and local distillation
η_t, C	Learning rate and number of classes in the dataset
K, S	Number of local training and distillation steps
$\phi_n(\cdot)$	Function characterising the impacts of overall distortion
$\phi_{n,DL}(\cdot)$	Function characterising the wireless impairments of DL
$\phi_{UL}(\cdot)$	Function characterising the wireless impairments of UL
N_{BS}, N_D	Number of antennas at BS and UE
$\mathbf{H}, \mathbf{F}, \mathbf{Z}, \hat{\mathbf{Z}}$	Matrics of CSI, precoder, logits and estimated logits
$\mathbf{\Gamma}^{UL}, \mathbf{\Gamma}^{DL}$	Matrics of received signal via UL and DL
$\mathbf{\Omega}^{UL}, \mathbf{\Omega}^{DL}$	Matrics of zero mean Gaussian noise for UL and DL
σ_z^2, σ^2	Variance of logits and Gaussian noise
λ_n	Weighting factor for user n
P_{UL}, P_{DL}	Transmit power for UL and DL communication

Notations: $\mathcal{B}$, $\mathbf{B}$, $\mathbf{b}$, b represent a set, a matrix, a vector, and a scalar, respectively. $\mathbf{B}^H$ and $\mathbf{B}^{-1}$ are the Hermitian transpose and the inverse of $\mathbf{B}$, respectively. tr($\mathbf{B}$) denotes the trace of $\mathbf{B}$. $\|\mathbf{b}\|_2$ is the l_2-norm of $\mathbf{b}$. $\mathcal{CN}(b, \mathbf{B})$ represents a circularly symmetric complex Gaussian random vector with mean b and covariance matrix $\mathbf{B}$. The notation $\mathbf{I}_K$ is the identity matrix of size $K \times K$. For an ordered set τ, $\tau(i)$ denotes i^{th} element in the set. Specifically, the notations used in this paper are summarized in Table II.

II. System Model

We consider a wireless single-cell multi-user cellular network with N users and a server or base station (BS). The BS has N_{BS} number of antennas, while each user device is equipped with N_D antennas.[1] The nth user stores a local private labeled dataset $\mathcal{D}_n = \{\mathbf{d}_i = (\mathbf{x}_i, \mathbf{y}_i), i = 1, \ldots, D_n\}$, that is, the total dataset across users $\mathcal{D} = \cup_{n=1}^{N} \mathcal{D}_n$ with size $D = \sum_{n=1}^{N} D_n$. Each data point $\mathbf{d}$ includes the input data $\mathbf{x}$ and the corresponding output label $\mathbf{y}$. We assume all users' datasets are not independent and identically distributed (non-i.i.d.). Similarly, $\mathcal{X} = \{\mathbf{x}_{p,i}, i = 1, \ldots, D_p\}$ denote the unlabeled public dataset known to both users and server.

The objective of users is to learn private heterogeneous models based on their local data. Let $\mathbf{z} = F_n(\mathbf{x}; \mathbf{w})$ denote the mapping with learning weights $\mathbf{w}$ from the input data point $\mathbf{x}$ to the output logits $\mathbf{z}$. The loss function between given labels ($\mathbf{y}$) and output logits of the model ($\mathbf{z}$) is assumed as $l(\mathbf{z}, \mathbf{y}) = l(F_n(\mathbf{x}; \mathbf{w}), \mathbf{y})$.[2] The overall loss function for the dataset $\mathcal{D}_n$ can be written as

$$L_n(\mathbf{w}) = \frac{1}{D_n} \sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}_n} l(F_n(\mathbf{x}; \mathbf{w}), \mathbf{y}). \tag{1}$$

The aim of the federated distillation can be formulated as

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} L_n(\mathbf{w}), \forall n. \tag{2}$$

When logits are available rather than labels such as in the distillation steps with the public dataset, Kullback-Leibler (KL) divergence is used for distillation loss function as [15]

$$Q_n(\mathbf{w}; \mathcal{Z}) = \frac{1}{|\mathcal{Z}|} \sum_{(\mathbf{x}_p, \mathbf{z}_p) \in \mathcal{X} \times \mathcal{Z}} l_{KL}\big(\mathrm{LSM}\big(F_n\big(\mathbf{x}_p; \mathbf{w}\big)\big), \mathrm{SM}\big(\mathbf{z}_p\big)\big), \tag{3}$$

where $\mathbf{z}_p \in \mathcal{Z}$ are the set of logits corresponding to the unlabeled public dataset of inputs $\mathbf{x}_p$. $|\mathcal{Z}|$ denotes the cardinality of set $\mathcal{Z}$. LSM and SM stand for log-softmax and softmax functions, respectively. In the above (3), model output logits are utilized, rather than labels, which is due to the fact that labels, the output of softmax layer, lie within the range (0, 1) and sum up to 1. Moreover, using model output logits provides more tolerance while communicating over imperfect links [24], [25], [27], [28].

A. Assumptions on the Loss Function

Following assumptions for both training loss function $L_n(\cdot), n = 1, \ldots, N$ and distillation loss function $Q_n(\cdot), n = 1, \ldots, N$, are made for the convergence analysis [30], [33], [34], [35].

Assumption 1 (Smoothness): $L_1, \ldots, L_N$ are all ψ-smooth, for $\forall \mathbf{v}, \mathbf{w}$, $L_n(\mathbf{v}) \leq L_n(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla L_n(\mathbf{w}) + \frac{\psi}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$.

Assumption 2 (Convexity): $L_1, \ldots, L_N$ are all μ-strongly convex, for $\forall \mathbf{v}, \mathbf{w}$, $L_n(\mathbf{v}) \geq L_n(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla L_n(\mathbf{w}) + \frac{\mu}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$.

$Q_1, \ldots, Q_N$ are all μ_q-strongly convex, for $\forall \mathbf{v}, \mathbf{w}$, $Q_n(\mathbf{v}) \geq Q_n(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla Q_n(\mathbf{w}) + \frac{\mu_q}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$.

Assumption 3 (Bounded Variance): The stochastic gradients for each client satisfy

$$\mathbb{E}\big\|\nabla L_n\big(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\big) - \nabla L_n\big(\mathbf{w}_{n,t}, \mathcal{D}_n\big)\big\|_2^2 \leq \sigma_n^2, \tag{4}$$

$$\mathbb{E}\big\|\nabla Q_n\big(\mathbf{w}_{n,t}, \mathcal{Z}_{n,t}\big) - \nabla Q_n\big(\mathbf{w}_{n,t}, \mathcal{Z}_n\big)\big\|_2^2 \leq \sigma_{qn}^2, \tag{5}$$

where $\mathcal{D}_{n,t} \subset \mathcal{D}_n$ is the mini-batch (a subset chosen uniformly and independently from n^{th} user's local training dataset $\mathcal{D}_n$ at time step t).

Assumption 4 (Uniformly Bounded Gradient): The squared l_2-norm of gradients satisfies

$$\mathbb{E}\big\|\nabla L_n\big(\mathbf{w}_{n,t}, \mathcal{D}_n\big)\big\|_2^2 \leq G^2, \forall n, \tag{6}$$

$$\mathbb{E}\big\|\nabla Q_n\big(\mathbf{w}_{n,t}, \mathcal{Z}_n\big)\big\|_2^2 \leq G_q^2, \forall n, \tag{7}$$

Assumptions 1 and 2 are the standard ones for l_2-norm regularized linear regression, logistic regression, and softmax

[1]Considering its implementation complexity and resulting training latency [31], [32], in the designed system, we do not include the error correction coding schemes. It should be noted that increasing the number of antennas at BS and user will provide system performance gain in a similar way.

[2]In this paper, we use cross entropy as local training loss function $l(\cdot)$.

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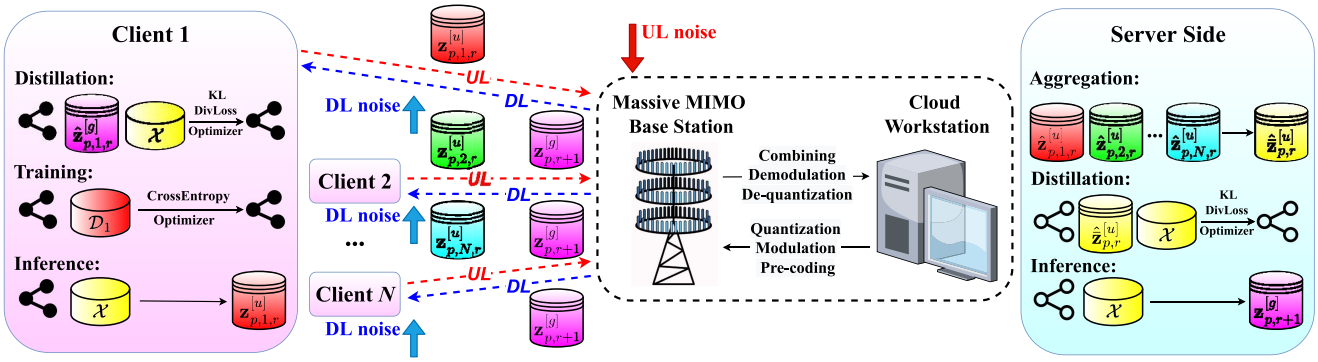

Fig. 1. Architecture of the proposed massive MIMO-integrated FD framework with imperfect communication links, where quantization noise, channel fading, intra-cell interference and receiver side noise are considered. $\mathcal{D}_1$ and $\mathcal{X}$ represent the personal dataset of client 1 and public dataset, respectively. $\mathbf{z}_{p,r+1}^{[g]}$ denotes the predicted logits on the public dataset by server side model at $(r+1)^{th}$ round and $\hat{\mathbf{z}}_{p,n,r+1}^{[g]}$ stands for its corresponding noisy version, received by n^{th} user. Note that all the local users share similar procedures as shown for Client 1, detailed local procedures for Client 2 and Client N are omitted in this figure for better visualization.

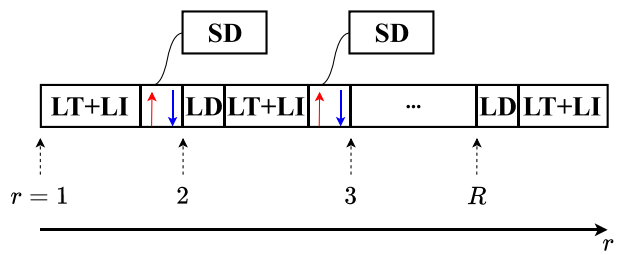

Fig. 2. Illustration of the timeline of communication rounds (r) in federated distillation. Up and down arrows signify the uplink and downlink communications, respectively. Abbreviations are defined as: "LT": Local user side Training, "LI": Local user side Inference, "LD": Local user side Distillation, "SD": Server side Distillation.

classifier. Assumptions 3 and 4 are based on bounded gradients and are helpful in bounding the model-error variance while analyzing the convergence.

B. Learning Dynamics

The process of FD learning is divided into three parts as shown in Fig. 1, viz, local training step, communication round, server and local distillation steps. The corresponding time slots are illustrated in Fig. 2. Let $\mathcal{T} = \{1, 2, \ldots, T\}$ denote the set for all time steps, that is,

$$\mathcal{T} = \mathcal{T}_{LT} \cup \mathcal{T}_{SD} \cup \mathcal{T}_{LD}, \tag{11}$$

where $\mathcal{T}_{LT}$, $\mathcal{T}_{SD}$ and $\mathcal{T}_{LD}$ denote the time indices for local training, server and local distillations, respectively;[3] K_r and S represent the numbers of local training and distillation steps in the rth round, respectively. As depicted in Fig. 2, the learning process starts with local training with K_r steps using stochastic gradient descent (SGD). The update equation for weights for each user can be expressed as

$$\mathbf{w}_{n,t+1} = \mathbf{w}_{n,t} - \eta_t \nabla L_n\big(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\big), t \in \mathcal{T}_{LT} \tag{12}$$

where η_t is the learning rate. After the private training process, logits are generated based on the public dataset $(\mathcal{X})$ as

$$\mathbf{z}_n = F_n\big(\mathbf{x}_p; \mathbf{w}_{n,t}\big), \forall \mathbf{x}_p \in \mathcal{X}, \tag{13}$$

and collected in the set $\mathcal{Z}_n$ for each $\mathbf{x}_p \in \mathcal{X}$. These logits are transmitted from each user via uplink noisy channels (red dashed lines in Fig. 1) to the server. The server aggregates and distills the logits received from all users in S steps. Then, the server sends the processed logits to the downlink noisy channels (blue dashed lines in Fig. 1). Let $\hat{\mathcal{Z}}_n = \{\hat{\mathbf{z}}_n\}$ denote the set of logits received at the nth user. The relation between $\mathcal{Z}$ and $\hat{\mathcal{Z}}_n$ is described in the next section. Subsequently, in the local distillation phase, the received logits are used to update the local models in S steps. Each distillation step can be given as

$$\mathbf{w}_{n,t+1} = \mathbf{w}_{n,t} - \eta_t \nabla Q_n\Big(\mathbf{w}_{n,t}, \hat{\mathcal{Z}}_{n,t}\Big), t \in \mathcal{T}_{LD}, \tag{14}$$

[3] In the set definition at the bottom of this page, short notation $a+(1,\ldots,k)$ for $\{a+1,\ldots,a+k\}$ is used.

$$\mathcal{T}_{LT} = \left\{\{1,\ldots,K_1\},\ldots,\left\{\sum_{i=1}^{r} K_i + 2rS + (1,\ldots,K_{r+1})\right\},\ldots\right\}, \tag{8}$$

$$\mathcal{T}_{SD} = \left\{\{K_1 + (1,\ldots,S)\},\ldots,\left\{\sum_{i=1}^{r+1} K_i + (1,\ldots,S)\right\},\ldots\right\}, \tag{9}$$

$$\mathcal{T}_{LD} = \left\{\{K_1 + S + (1,\ldots,S)\},\ldots,\left\{\sum_{i=1}^{r+1} K_i + S + (1,\ldots,S)\right\},\ldots\right\}, \tag{10}$$

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where $\hat{\mathcal{Z}}_{n,t} \subset \hat{\mathcal{Z}}_n$ is the mini-batch. This distillation step is followed by similar local training steps and so on.

III. Uplink and Downlink Communications

Since BS is equipped with a large number of antennas, we consider ZF approach at the BS, and analyze the impacts on model outputs using channel hardening effect. Performance of ZF techniques in massive MIMO system has been investigated in [36] and [37].

In the following subsections, we analyze the error introduced by the massive MIMO communication for the transmitted logits, which are the information exchanged between local users and BS. Let the function $\phi_n(\cdot)$ represent the distortion arising due to uplink and downlink communications, which can be decomposed into uplink operation $\phi_{UL}(\cdot)$, server aggregation and distillation $\phi_{av}(\cdot)$, and downlink broadcast operation $\phi_{n,DL}(\cdot)$, i.e., a composition of three mappings as $\phi_n(\mathbf{z}_{n,t}, \forall n) = \phi_{n,DL} \circ \phi_{av} \circ \phi_{UL}(\mathbf{z}_{n,t}, \forall n)$.

A. Uplink Mapping

The uplink process first involves resizing $D_p \times C$ real-valued logits $\mathbf{z}_{n,t}$ into complex-valued data streams for transmission, say $\mathbf{Z}_{n,t} \in \mathbb{C}^{N_D \times \tau_{UL}}$, where $\tau_{UL} = \frac{D_p C}{2N_D}$. The received signal at the massive MIMO BS can be written as

$$\mathbf{\Gamma}_t^{(UL)} = \sum_{n=1}^{N} \mathbf{H}_t^n \mathbf{Z}_{n,t} \sqrt{\frac{P_{UL}}{N_D \sigma_z^2}} + \mathbf{\Omega}_t^{(UL)}, \tag{15}$$

where entries of $\mathbf{Z}_{n,t}$ are assumed to have zero mean and σ_z^2 variance; $\mathbf{\Omega}_t$ denotes zero-mean Gaussian noise with σ^2 variance, and P_{UL} is the transmit power constraint for each device; that is, $tr\mathbb{E}\{\mathbf{Z}_{n,t}\mathbf{Z}_{n,t}^H\} = \tau_{UL} N_D \sigma_z^2$. With the zero forcing receiver with $N_{BS} \geq N \cdot N_D$ number of antennas, the estimated logits at the BS can be given as

$$\begin{aligned}\hat{\mathbf{Z}}_t &= \sqrt{\frac{N_D \sigma_z^2}{P_{UL}}} \left(\mathbf{H}_t^H \mathbf{H}_t\right)^{-1} \mathbf{H}_t^H \mathbf{\Gamma}_t^{(UL)} \\ &= \mathbf{Z}_t + \sqrt{\frac{N_D \sigma_z^2}{P_{UL}}} \left(\mathbf{H}_t^H \mathbf{H}_t\right)^{-1} \mathbf{H}_t^H \mathbf{\Omega}_t^{(UL)},\end{aligned} \tag{16}$$

where $\mathbf{H}_t = [\mathbf{H}_t^1, \ldots, \mathbf{H}_t^N]$ and $\mathbf{Z}_t^T = [\mathbf{Z}_{1,t}^T, \ldots, \mathbf{Z}_{N,t}^T]$. Each column of the estimation error term has zero mean and covariance matrix as $\frac{N_D \sigma_z^2 \sigma^2}{P_{UL}} (\mathbf{H}_t^H \mathbf{H}_t)^{-1}$. Owing to channel hardening $\frac{1}{N_{BS}} \mathbf{H}_t^H \mathbf{H}_t \rightarrow \mathbf{I}$, the covariance matrix of the effective noise above can be simplified as $\frac{N_D \sigma_z^2 \sigma^2}{P_{UL}} N_{BS}^{-1} \mathbf{I}_{N_D}$. The variable $\hat{\mathbf{Z}}_t$ can be used to express the estimated model outputs for each user at the BS as $\hat{\mathbf{z}}_{n,t} = \mathbf{z}_{n,t} + \omega_{n,t,UL}$, where $\omega_{n,t,UL} \sim \mathcal{N}(\mathbf{0}, \frac{N_D \sigma_z^2 \sigma^2}{2P_{UL} N_{BS}} \mathbf{I})$ and the factor of 2 in the denominator comes from converting the complex random variables to the real ones.

B. Server Aggregation and Distillation Mapping

We assume server distillation of logits from all users is similar to aggregation with an uncertainty ω_D, defined by a Gaussian distribution having zero mean and small variance σ_D^2. Therefore, having $\hat{\mathbf{Z}}_t$ at the BS, the averaging and distillation operation can be given as

$$\begin{aligned}\phi_{av}\left(\hat{\mathbf{Z}}_t\right) &= \sum_{n=1}^{N} \lambda_n \hat{\mathbf{z}}_{n,t} + \omega_D && (17)\\ &= \bar{\mathbf{z}}_t + \bar{\omega}_{t,UL} + \mathbf{\Omega}_D && (18)\\ &= \bar{\mathbf{z}}_t + \bar{\omega}_{t,av}, && (19)\end{aligned}$$

where $\bar{\mathbf{z}}_t = \sum_{n=1}^{N} \lambda_n \mathbf{z}_{n,t}$, $\lambda_n = \frac{D_n}{\sum_{n=1}^{N} D_n}$ is the weighting factor for n^{th} user and $\|\boldsymbol{\lambda}\|_2^2 = \sum_{n=1}^{N} \lambda_n^2$. $\bar{\omega}_{t,UL} \sim \mathcal{N}(\mathbf{0}, \frac{\|\boldsymbol{\lambda}\|_2^2 \sigma_z^2}{2} \frac{\sigma^2 N_D}{P_{UL} N_{BS}} \mathbf{I})$ and $\bar{\omega}_{t,av} \sim \mathcal{N}(\mathbf{0}, [\frac{\|\boldsymbol{\lambda}\|_2^2 \sigma_z^2}{2} \frac{\sigma^2 N_D}{P_{UL} N_{BS}} + \sigma_D^2]\mathbf{I})$

C. Downlink Mapping

The aggregated vector, say $\hat{\bar{\mathbf{z}}}_t = \phi_{av}(\hat{\mathbf{Z}}_t)$, is then broadcasted to each user device via the base station through the DL channel. Since each device can receive only N_D data streams, the vector $\hat{\bar{\mathbf{z}}}_t$ is resized into a complex-valued matrix $\hat{\bar{\mathbf{Z}}}_t \in \mathbb{C}^{N_D \times \tau_{DL}}$ with $\tau_{DL} = \frac{D_p C}{2N_D}$. The received signal at the nth device can be given as

$$\mathbf{\Gamma}_{n,t}^{(DL)} = \mathbf{H}_t^{nH} \mathbf{F}_t \hat{\bar{\mathbf{Z}}}_t p_{DL} + \mathbf{\Omega}_{n,t}^{(DL)}, \tag{20}$$

where channel reciprocity is assumed; and $\mathbf{F}_t$ denotes the $N_{BS} \times N_D$ precoder. For ZF scheme, $\mathbf{F}_t = \frac{1}{\sqrt{N_{BS}}} \mathbf{H}_t^n (\mathbf{H}_t^{nH} \mathbf{H}_t^n)^{-1}$. The transmit power constraint (P_{DL}) is given as

$$p_{DL}^2 = \frac{P_{DL} \tau_{DL}}{tr\mathbb{E}\left\{\mathbf{F}_t \hat{\bar{\mathbf{Z}}}_t \hat{\bar{\mathbf{Z}}}_t^H \mathbf{F}_t^H\right\}} = \frac{P_{DL}}{\hat{\sigma}_z^2 N_D}, \tag{21}$$

where $\hat{\sigma}_z^2 = \|\boldsymbol{\lambda}\|_2^2 \sigma_z^2 (1 + \frac{\sigma^2 N_D}{P_{UL} N_{BS}}) + \sigma_D^2$. Then, the estimation at the nth user can be obtained as $(\mathbf{H}_t^{nH} \mathbf{F}_t)^{-1} \mathbf{\Gamma}_{n,t}^{(DL)} p_{DL}^{-1}$. The corresponding covariance matrix for each column of the estimation error $(\mathbf{H}_t^{nH} \mathbf{F}_t)^{-1} \mathbf{\Omega}_{n,t}^{(DL)} p_{DL}^{-1}$ can be computed as

$$\begin{aligned}&\frac{\hat{\sigma}_z^2 N_D \sigma^2}{P_{DL}} \left(\mathbf{H}_t^{nH} \mathbf{F}_t\right)^{-1} \left(\mathbf{H}_t^{nH} \mathbf{F}_t\right)^{-H} \\ &= \frac{\hat{\sigma}_z^2 N_D \sigma^2}{P_{DL}} \left(\mathbf{F}_t^H \mathbf{H}_t^n \mathbf{H}_t^{nH} \mathbf{F}_t\right)^{-1} \approx \frac{\hat{\sigma}_z^2 \sigma^2}{P_{DL}} \mathbf{I},\end{aligned} \tag{22}$$

where $\mathbf{H}_t^n \mathbf{H}_t^{nH}$ is approximated to $N_D \mathbf{I}_{N_{BS}}$, owing to channel hardening effect. Thus, we can write $\phi_{n,DL}(\hat{\bar{\mathbf{z}}}_t) = \hat{\bar{\mathbf{z}}}_t + \bar{\omega}_{n,t,DL}$, where $\bar{\omega}_{n,t,DL} \sim \mathcal{N}(\mathbf{0}, \frac{\hat{\sigma}_z^2 \sigma^2}{2P_{DL}} \mathbf{I})$ is the Gaussian noise at the typical user.

Therefore, the composite uplink-downlink aggregation process can be expressed as

$$\begin{aligned}\phi_n\left(\mathbf{z}_{n,t}, \forall n\right) &= \phi_{n,DL} \circ \phi_{av} \circ \phi_{UL}\left(\mathbf{z}_{n,t}, \forall n\right) && (23a)\\ &= \phi_{n,DL} \circ \phi_{av}\left(\hat{\mathbf{Z}}_t\right) && (23b)\\ &= \phi_{n,DL}\left(\bar{\mathbf{z}}_t + \bar{\omega}_{t,av}\right) && (23c)\\ &= \bar{\mathbf{z}}_t + \underbrace{\bar{\omega}_{t,av} + \bar{\omega}_{n,t,DL}}_{\bar{\omega}_{n,t}}, && (23d)\end{aligned}$$

where $\bar{\mathbf{z}}_t = \sum_{n=1}^{N} \lambda_n \mathbf{z}_{n,t}$, and proof of the variance for the effective noise $\bar{\omega}_{n,t}$, σ_ω^2, can be seen in Appendix A.

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Remark: In the above mapping, the weights can be considered with quantization and modulation by introducing quantization error, e.g., with uniform quantization for real and imaginary parts in Eqn. (15), the resulting error variance can be given as sum of noise variance σ^2 and the variance of the quantization error $\frac{\Delta^2}{6}$, where Δ is the quantization step size. The total effective variance can be obtained similarly with additive quantization error variance, which affects the convergence in the same way as the noise variance σ_ω^2.

IV. Convergence Analysis

In this section, we present our convergence analysis for the FD paradigm, where logits are treated as the only information exchanged between the server and local users. We consider the scenario where weights are updated at the end of each time slot.

Although the convergence analysis in [19] also has a similar objective function, the upper bound is derived with respect to data distribution, number of data samples and users, not for the time step t as we have done in this work. On the other hand, the convergence analysis provided in work [20] only includes the distillation module, not the whole algorithm.

The key idea of our proof is to consider both training and distillation processes. First, we upper-bound the next step error in Lemma 1 and Lemma 2 by utilizing the assumptions discussed in Section II. Further, for both cases, with the help of induction, we bound the term of $\mathbb{E}\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\|_2^2$ and $\mathbb{E}\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\|_2^2$ in a unified format of $\mathcal{O}(t^{-1})$, as will be shown in Corollary 1 and Corollary 2, respectively. And these two results will be used jointly to derive our main Theorem 1, which bounds the values of the global loss function with respect to the time steps.

We now begin to present our convergence analysis. First, identifying the fact that in the server side distillation process, local model weights are not updated, we have, $\mathbf{w}_{n,t} = \mathbf{w}_{n,t-S}, t = \tau(S), \tau \in \mathcal{T}_{SD}$.

Therefore, considering the unchanged local weights during the server distillation process, we can write the relation of the weights during server and local distillation rounds as

$$\mathbf{w}_{n,t+2S} = \mathbf{w}_{n,t} - \sum_{i=0}^{S-1} \eta_{t+S+i} \nabla Q_n\Big(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\Big),$$

where $t = \tau(K_r), \tau \in \mathcal{T}_{LT}$.

In the following lemma, bounded variance and bounded gradient form Assumption 3 and 4 are utilized to bound the next step error during the local training process.

Lemma 1 (Next Step Update for Local Training Process): Let Assumption 1-4 hold. Then, when $t+1 \in \mathcal{T}_{LT}$, with respect to the optimized model weights $\mathbf{w}_n^*$, the squared l_2-norm of the next step model update error can be bounded as

$$\mathbb{E}\big\|\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\big\|_2^2 \le (1 - \eta_t \mu)\mathbb{E}\big\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\big\|_2^2 + \eta_t^2\Big(\sigma_n^2 + G^2\Big). \tag{24}$$

Proof: Proof is provided in Appendix B. ■

In addition, we can further bound the term $\mathbb{E}\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\|_2^2$ with respect to time step t as follows (proof will be provided in the appendix), and this result will be used when we derive the upper bound of the modeling error for the proposed FD algorithm.

Corollary 1: From Lemma 1, the term $\mathbb{E}\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\|_2^2$ can be further bounded as

$$\mathbb{E}\big\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\big\|_2^2 \le \frac{\nu_n}{t + \beta_1} \tag{25}$$

where $\nu_n \ge \frac{\beta_0^2 C_n}{\beta_0 \mu - 1}, C_n = (\sigma_n^2 + G^2), \eta_t = \frac{\beta_0}{t+\beta_1}, \beta_0 > \frac{1}{\mu}, \beta_1 > 0$.

Proof: Proof is derived in Appendix C. ■

Similarly, for the server and local distillation processes, we have the following bound.

Lemma 2 (Next Step Update for Server and Local Distillation Processes): Let Assumption 1-4 hold. Then, at the end of local training, when $t = \tau(K_r), \tau \in \mathcal{T}_{LT}$, with respect to the optimized model weights $\mathbf{w}_n^*$, the squared l_2-norm of the next step model update error can be bounded as

$$\begin{aligned}&\mathbb{E}\big\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\big\|_2^2 \\ &\le \left(1 - \mu_q \sum_{i=0}^{S-1} \eta_{t+S+i}\right)\mathbb{E}\big\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\big\|_2^2 \\ &\quad + \left(\sum_{i=0}^{S-1} \eta_{t+S+i}^2\right) S\Big(\sigma_{qn}^2 + G_q^2\Big).\end{aligned} \tag{26}$$

Proof: Proof is provided in Appendix D. ■

Corollary 2: From Lemma 2, the term $\mathbb{E}\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\|_2^2$ can be further bounded as

$$\mathbb{E}\big\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\big\|_2^2 \le \frac{\nu_n}{t + 2S + \beta_1} \tag{27}$$

where $\nu_n \ge \frac{S^2 \beta_0^2 C_{qn}}{\beta_0 \mu_q - 2S}, C_{qn} = (\sigma_{qn}^2 + G_q^2), \eta_t = \frac{\beta_0}{t+\beta_1}, \beta_0 > \frac{2S}{\mu_q}, \beta_1 \ge 4$.

Proof: Proof is derived in Appendix E. ■

The above results indicate that if μ, μ_q and η_t are chosen properly, one can show the model updates lead to non-increasing errors; that is, the convergence of the proposed FD algorithm is guaranteed. The results of the above lemmas are aggregated and used in the following theorem, which employs the smoothness of loss functions (Assumption 1).

Theorem 1: Let Assumption 1-4 hold. Then, the proposed FD algorithm satisfies

$$\mathbb{E}\big\{L\big(\mathbf{w}_{n,t+1}\big)\big\} - L_n^* \le \frac{\psi}{2} \cdot \frac{\nu_n}{t + 1 + \beta_1}, \tag{28}$$

where $\nu_n \ge \max(\frac{\beta_0^2 C_n}{\beta_0 \mu - 1}, \frac{S^2 \beta_0^2 C_{qn}}{\beta_0 \mu_q - 2S}), C_{qn} = (\sigma_{qn}^2 + G_q^2), C_n = (\sigma_n^2 + G^2), \beta_0 > \max(\frac{1}{\mu}, \frac{2S}{\mu_q}), \beta_1 \ge 4$.

Proof: Proof is provided in Appendix F. ■

Theorem 1 tells that in the proposed FD algorithm, modeling error reduces in $\mathcal{O}(t^{-1})$ manner. The derived convergence bound relies on the values of ψ, μ, C_n and C_{qn}, which are from the statistical properties of L_n and Q_n; that is, the loss functions should be selected carefully. The convergence bound linearly depends on the ψ-smoothness, whereas it is inversely proportional to the μ-convexity; that is, the stronger the convexity is, the lower the convergence

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Algorithm 1 Dynamic Training Steps Based FD Algorithm (FedTSKD)

1: **for** $r = 1, \ldots, R$ **do**
2: **if** $r \neq 1$ **then**
3: Downlink transmission:
4: 1. BS quantizes, modulates and precodes logits.
5: 2. BS broadcasts $\mathbf{z}_{p,r}^{[g]}$ to all users.
6: Local distillation:
7: 1. Local users apply softmax function to the received logits.
8: 2. Local users perform local distillation.
9: **end if**
10: **for** $n = 1, \ldots, N$ **do**
11: **for** $t = \mathcal{T}_{LT}(r)$ **do**
12: Update local weights as:
13: $\mathbf{w}_{n,t+1} = \mathbf{w}_{n,t} - \eta_t \nabla L_n(\mathbf{w}_{n,t}, \mathcal{D}_{n,t})$
14: **end for**
15: n^{th} user performs inference on the public dataset and gets corresponding logits $\mathbf{z}_{p,n,r}^{[u]}$.
16: **end for**
17: Uplink transmission:
18: 1. Local users quantize and modulate logits.
19: 2. Users upload logits to server.
20: 3. BS combines, demodulates and dequantizes the received logits.
21: Logits aggregation at BS:
22: 1. BS aggregates the logits sent from users to produce $\hat{\tilde{\mathbf{z}}}_{p,r}^{[u]}$.
23: 2. $\hat{\tilde{\mathbf{z}}}_{p,r}^{[u]}$ is passed through softmax function.
24: Server side model performs distillation.
25: Server side model predicts on the public dataset and obtains $\mathbf{z}_{p,r+1}^{[g]}$.
26: **end for**

error is. Further, from the above theorem, it can be concluded that the loss value at the convergence (ν_n) depends on S, the number of distillation steps. When S, L_n and Q_n are selected such that $\nu_n \geq \frac{S^2 \beta_0^2 C_{qn}}{\beta_0 \mu_q - 2S}$ and $\beta_0 > \frac{2S}{\mu_q}$, for a given time step t, the modeling error convergence bound increases in $\mathcal{O}(S^3)$ manner. On the other hand, when $\nu_n \geq \frac{\beta_0^2 C_n}{\beta_0 \mu - 1}$ and $\beta_0 > \frac{2S}{\mu_q}$, we get $\mathcal{O}(S)$ bound. For $\beta_0 > \frac{1}{\mu}$, if $\nu_n \geq \frac{\beta_0^2 C_n}{\beta_0 \mu - 1}$, the bound is independent of S. This is an interesting finding since this implies that the loss functions for local training and distillation could be dynamically changed during the training process to accelerate the model convergence and performance. However, due to limited space, we would like to delegate further study to our future investigation.

A. Proposed Dynamic Training Steps Based FD Algorithm (FedTSKD)

Algorithm 1 summarizes the proposed massive MIMO-integrated training steps varied FD algorithm (FedTSKD), where for each round (r), only the uplink transmission takes place after the local updates. Lines 3-5 denote the downlink transmission from BS to users, where quantization, modulation and precoding are performed for distilled logits. After getting logits at the user side, users process it locally using softmax-function and distillation as shown in lines 6-8. These logits are further updated via local training steps simultaneously for each user, where the number of steps, denoted by $\mathcal{T}_{LT}(r)$, can differ each round, and can be dynamically chosen based on the training progress history. Subsequently, to prepare for federated transmission, logits are obtained by operating on the public dataset. These logits are quantized and modulated. The received logits at the BS are aggregated and processed using softmax function. After BS's distillation, the public dataset's logits are obtained again for the downlink process, given in lines 3-5.

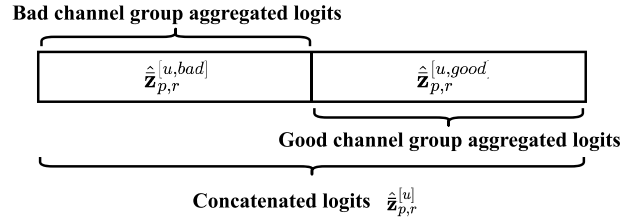

Fig. 3. Concatenation of logits at server side.

It can be noted that in order to obtain and improve communication-efficiency in the proposed FD algorithm, we decrease the number of local training steps as the rounds proceed (shown in lines 11-14), i.e., $K_1 \geq K_2 \geq \cdots \geq K_r \geq \cdots \geq K_R$. It can be seen from simulations that near convergence, reducing local iterations cannot further improve the performance, e.g., $K_r = K_{r+1}$.

V. Communication Channel-Aware FD Algorithm

In this section, we propose group-based communication channel-aware FD algorithm, where the FD system is made adaptive to the different communication links. Further, Algorithm 1 is improvised including these changes.

A. Group-Based FD Algorithm (FedTSKD-G)

For the communication channel-aware FD framework, the main design idea is to include the effect of channel quality within the distillation process and let the model learn different channel conditions intrinsically at server side. The received logits information from users with different channel strengths are processed separately.

In wireless communications, due to the reflection, diffraction, and scattering effects caused by the obstacles, when transmitted through wireless medium, the attenuation experienced by different signals from various transmitters/receivers will vary, leading to inconsistent channel quality among users. Specifically, two channel qualities, namely "good" and "bad" are considered as shown in the Fig. 3.

These different channel strengths can degrade the logits communications, and in turn can deteriorate the performance of the FD system, since in the local training steps and in distillation, the UL and DL errors are propagated. Note that at the BS, logits from users of good and bad channel strengths

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Algorithm 2 Group-Based FD Algorithm (FedTSKD-G)

1: **for** $r = 1, \ldots, R$ **do**
2: **if** $r \neq 1$ **then**
3: Downlink transmission:
4: 1. BS splits the predicted logits into two groups.
5: 2. BS quantizes, modulates and precodes logits.
6: 3. BS broadcasts $\mathbf{z}_{p,r}^{[g,bad]}$ and $\mathbf{z}_{p,r}^{[g,good]}$ to target users.
7: Local distillation:
8: 1. Local users apply softmax function to the received logits.
9: 2. Local users perform local distillation.
10: **end if**
11: **for** $n = 1, \ldots, N$ **do**
12: **for** $t = \mathcal{T}_{LT}(r)$ **do**
13: Update local weights as:
14: $$\mathbf{w}_{n,t+1} = \mathbf{w}_{n,t} - \eta_t \nabla L_n(\mathbf{w}_{n,t}, \mathcal{D}_{n,t})$$
15: **end for**
16: n^{th} user performs inference on the public dataset and gets corresponding logits $\mathbf{z}_{p,n,r}^{[u]}$.
17: **end for**
18: Uplink transmission:
19: 1. Local users quantize and modulate logits.
20: 2. Users upload logits to server.
21: 3. BS combines, demodulates and dequantizes the received logits.
22: Logits manipulation at BS:
23: 1. BS aggregates the logits sent from bad and good channel groups, respectively.
24: 2. BS concatenates the aggregated logits from these two groups and obtains $\hat{\hat{\mathbf{z}}}_{p,r}^{[u]}$.
25: 3. $\hat{\hat{\mathbf{z}}}_{p,r}^{[u]}$ is passed through softmax function.
26: Server side model performs distillation.
27: Server side model predicts on the public dataset and obtains $\mathbf{z}_{p,r+1}^{[g]}$.
28: **end for**

are aggregated/merged together. It means users with relatively good channels will also suffer, leading to slower convergence with worse testing accuracy.

To tackle this unfair aggregation, received logits at BS from two groups of users (good and bad) are aggregated separately, providing $\hat{\hat{\mathbf{z}}}_{p,r}^{[u,bad]}$ and $\hat{\hat{\mathbf{z}}}_{p,r}^{[u,good]}$. These steps are included in the Algorithm 2, which is improvised over Algorithm 1. These aggregated logits are then concatenated for distillation (see lines 22-26), that is, $\hat{\hat{\mathbf{z}}}_{p,r}^{[u]} = [\hat{\hat{\mathbf{z}}}_{p,r}^{[u,bad]}, \hat{\hat{\mathbf{z}}}_{p,r}^{[u,good]}]$. Accordingly, the output dimension of the last fully connected layer of the server-side model is doubled. In this way, the server-side model learns the influences of two different channel qualities on received logits. After distillation, predicted logits $\mathbf{z}_{p,r}^{[g]}$ of the public dataset are split into two groups ($\mathbf{z}_{p,r}^{[g,bad]}$ and $\mathbf{z}_{p,r}^{[g,good]}$), and broadcasted to target users via downlink transmission. In other words, users will receive different updates of logits, based on their channel qualities.

VI. Simulation Results

In the simulation setup, each user learns a convolutional neural network (CNN) model for image classification task on private datasets ImageNette [38], CIFAR-10 [39] and MNIST [40], where for federated distillation, unlabeled STL-10 [41] and FMNIST [42] are used as the public datasets. CIFAR-10 and STL-10 datasets contain 10 classes of natural color images with sizes 32×32 and 96×96, whereas MNIST dataset includes handwriting number images 0 to 9 of size 28×28, and fashion items in FMNIST dataset consists of gray-scale images of size 28×28 with 10 classes. ImageNette is a subset of ImageNet [43] with 10 classes, and the average image resolution is 469×387.

Number of users N is set to be 15, each with single antenna $N_D = 1$; number of global communication rounds R is 200; the BS has $N_{BS} = 64$ antennas. For i.i.d. multi-path Rayleigh fading model, Gaussian noise is assumed to have zero mean and unit variance at both local user and BS sides. Uniform quantization scheme with 8 bits is employed to quantize the predicted logits, followed by Quadrature Phase Shift Keying (QPSK) for modulation. To obtain two groups of users, a path loss factor (0.3) is multiplied with CSI strength to simulate the channels with relatively bad quality. Full participation is considered; that is, all local users join the training process for each global communication round, since the communication payload required in the FD algorithm is much less than that in the FL scheme.

CIFAR-10 + STL-10 datasets: For private datasets of users, 50,000 samples of CIFAR-10 dataset are split in non-i.i.d. manner using Dirichlet distribution [44] with the parameter α enabling us to control the degree of non-i.i.d.-ness. Default value of α is 0.5, unless stated otherwise. A lower value of α implies a higher degree of non-i.i.d.-ness. Note that the testing sets have i.i.d. data samples. For the unlabeled public dataset, which is common among users, a set with 2000 images from STL-10 dataset is taken. The Adam optimizer [45] is used with learning rate $\eta = 0.001$ for both distillation and training. Batch sizes for training and distillation are set to be 128 and 500, respectively. For data augmentation, each sample is randomly flipped horizontally with probability of 0.5. Number of epochs is set to 2 for both distillation and training. When the number of training steps is varied, for the first 100 global communication rounds, we set $K_r = \frac{D_n}{|\mathcal{D}_{n,t}|}(5 - \lfloor \frac{r-1}{25} \rfloor)$. To analyze the models' heterogeneity provided by FD, local model at each user is chosen from one of three architectures ResNet [46], MobileNet [47] and ShuffleNet [48] CNNs.

ImageNette + STL-10 datasets: For the ImageNette dataset, we consider i.i.d. private dataset distribution and 9469 training images are apportioned among users. All other parameters and configurations in our simulation align with those employed in the CIFAR-10 + STL-10 scenario.

MNIST + FMNIST datasets: In the MNIST dataset, 3000 sample images are divided among N users for non-i.i.d. private datasets using Dirichlet distribution. Uniform random rotation (ranging from -20 degree to 20 degree), random horizontal translation (ranging from -3 pixels to 3 pixels) and random vertical translation (ranging from -3

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TABLE III
MODEL ARCHITECTURES

CNN_1	CNN_2	CNN_3
Conv 1-32	Conv 1-12	Conv 1-8
Maxpool	Maxpool	Maxpool
Conv 32-64	Conv 12-24	Conv 8-16
Maxpool	Maxpool	Maxpool
FC-128	FC-64	FC-64
FC-10	FC-10	FC-10
Dropout	Dropout	Dropout
1,199,882	78,714	25,034

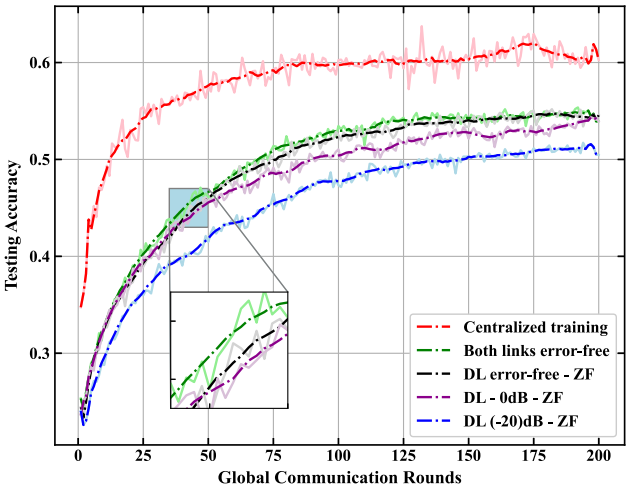

Fig. 4. Averaged testing accuracy across users with CIFAR-10/STL-10 datasets with different DL SNRs and fixed UL SNR -8 dB.

pixels to 3 pixels) are performed on each training image for data augmentation. Batch size is set to be 20 for training and 500 for distillation. For distillation, a public dataset is constructed with 2000 images from FMNIST dataset. Local user models follow a standard conventional CNN architecture which includes several stacks of convolutional layer, ReLU activation layer and max-pooling layer. For model heterogeneity, model structures are varied among users and detailed model configurations can be found in Table III,[4] where the values in the last row denote the total number of trainable parameters for each model.

In the following simulation results, for better visualization of the results, moving average (with a window size of 10) is plotted along with the actual values using the corresponding transparent colors. We use CUDA version 11 and Pytorch 1.7 [49] as libraries for training the models. Simulations are run on the system with one Intel i9-10980XE CPU (3 GHz) and one NVIDIA GeForce RTX 3090 GPU (24 GB).

A. Error-Prone and Error-Free Comparison

In Fig. 4, the testing accuracy with different values of DL SNR is plotted for a fixed UL SNR -8 dB. Green and black curves represent two baselines with error-free UL/DL links [15], [16], [17], [18], [19], [20], [21], [22] and error-free DL link [24], respectively. It can be observed that for both error-free and error-prone links, the testing accuracies converge; however, with more error-prone links, the accuracy achieved is lower and takes longer to converge. With -20 dB DL SNR, the achieved accuracy is approximately 91% of the accuracy for error-free links. Red curve stands for the centralized training baseline, where a single node contains all the training samples.

[4]Each user maintains one of these three CNN models locally. For all the CNNs, the convolutional layers contain kernels of size 3×3 and are followed by a ReLU activation layer.

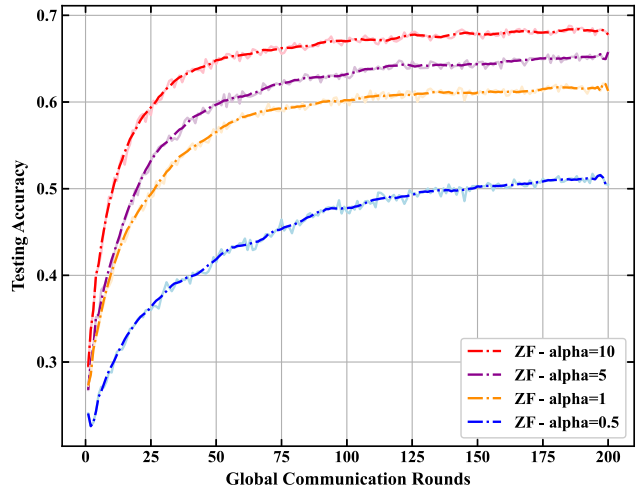

Fig. 5. Averaged testing accuracy across users with CIFAR-10/STL-10 datasets for UL SNR -8 dB, DL SNR -20 dB, and different non-i.i.d.-ness $\alpha = 0.5, 1, 5, 10$.

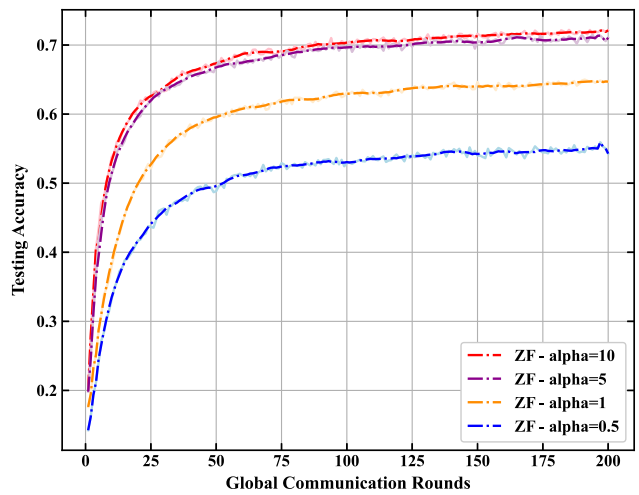

Fig. 6. Averaged testing accuracy with MNIST/FMNIST datasets for UL SNR -8 dB, DL SNR -20 dB, and different non-i.i.d.-ness $\alpha = 0.5, 1, 5, 10$.

B. Different Degrees of Non-i.i.d.-ness (α)

In Fig. 5, the value of non-i.i.d. factor α is varied and the progress of averaged testing accuracy is plotted for UL SNR -8 dB and DL SNR -20 dB, respectively. It can be seen that the testing accuracy is higher as α is increased, that is, non-i.i.d.-ness is decreased. In other words, the closer samples are to i.i.d., the higher is the accuracy.

In Fig. 6, effects of non-i.i.d.-ness variations are presented for MNIST/FMNIST datasets for fixed UL and DL SNRs -8

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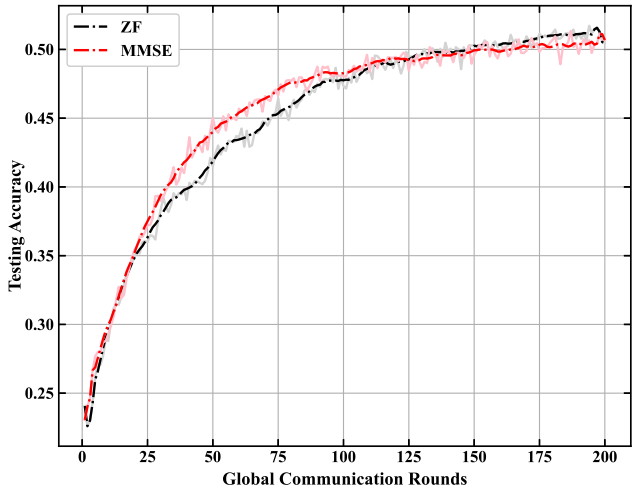

Fig. 7. Averaged testing accuracy across users for different precoding methods at fixed UL-SNR −8 dB and DL-SNR −20 dB with CIFAR-10/STL-10 datasets.

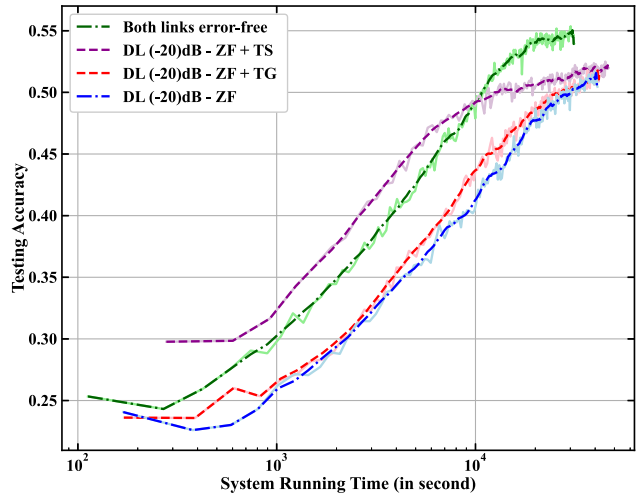

Fig. 8. Averaged testing accuracy across users for CIFAR-10/STL-10 datasets with fixed UL-SNR −8 dB and DL-SNR −20 dB (TG: Two Group Split, TS: Training Steps Variations).

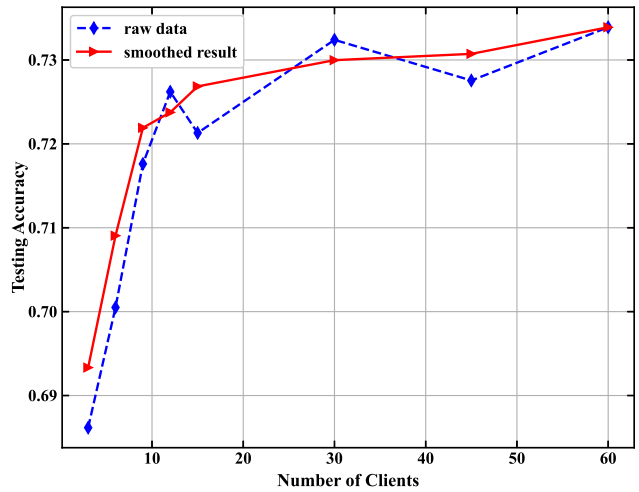

Fig. 9. Converged testing accuracy with MNIST/FMNIST datasets when different numbers of clients are considered. UL SNR −8 dB, DL SNR −20 dB, $\alpha = 10$.

and −20 dBs, respectively. Compared with the result presented in Fig. 5 for CIFAR-10/STL-10 datasets, with same degree of non-i.i.d.-ness, increments in the testing accuracy can be witnessed in Fig. 6, although in this scenario, each local user has much fewer training samples. This is due to the fact that MNIST dataset consists of single channel gray-scale image samples, whereas CIFAR-10 dataset has three-channel RGB images, making it more difficult to learn than the dataset with gray-scale images.

C. With Different Precoding Methods

In Fig. 7, the averaged testing accuracy is plotted for non-i.i.d. CIFAR-10/STL-10 datasets with ZF and MMSE precoding/combining methods. It can be witnessed that, both ZF and MMSE precoding strategies achieve similar performance; however at lower rounds, MMSE provides better accuracy. In high SNR regime, this difference becomes negligible, therefore, for the rest of simulations, we adopt the ZF strategy.

D. With Training Steps Variations and Group Splitting

In Fig. 8, the averaged testing accuracy versus CPU running time for different variations of the FD algorithm with the same number of communication rounds is plotted. It can be seen that the converged accuracy is higher if links are error-free (green curve). The blue curve depicts the error-prone accuracy without any variations with finite DL/UL SNRs. To make the algorithm converge faster with error-prone UL/DL links, number of training steps per round is varied (TS-labeled purple curve). In this variation, the numbers of training steps at earlier rounds are higher than those in later rounds, leading to converging faster and requiring fewer communication resources. To reduce the effect of communication errors at users, two group split scheme (TG-labeled red curve, without TS) shows better testing accuracy at earlier rounds than that of the blue curve, and both incur similar running time.

E. Model Heterogeneity

In Table IV, we summarize the averaged testing accuracies for user groups with nonidentical model structures when both error-free and error-prone channels (UL SNR −8 dB and DL SNR −20 dB) are considered, and different degrees of non-i.i.d.-ness (α) are utilized. It is worth noting that, for the scenario with CIFAR-10/STL-10 datasets, compared with ResNet and MobileNet groups, it can be witnessed that users with the simplest architecture ShuffleNet have the most unsatisfactory performance (an approximate performance degradation of 6%, 5% and 8% when $\alpha = 0.5$, 1 and 5, respectively). This implies that achieving successful convergence of the FD algorithm requires the simultaneous consideration of both data heterogeneity and model complexity.

F. With Different Numbers of Clients

In Fig. 9, the averaged testing accuracies are plotted with respect to different numbers of clients. MNIST is used as

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TABLE IV
CONVERGED TESTING ACCURACIES FOR USER GROUPS WITH NONIDENTICAL MODEL ARCHITECTURES WHEN ERROR-FREE AND ERROR-PRONE CHANNELS ARE CONSIDERED WITH DIFFERENT DEGREES OF NON-I.I.D.-NESS (α). TWO DIFFERENT DATASET COMBINATIONS ARE UTILIZED

		CIFAR-10+STL-10 datasets				MNIST+FMNIST datasets		
Settings	Model group	$\alpha = 0.5$	$\alpha = 1$	$\alpha = 5$	Model group	$\alpha = 0.5$	$\alpha = 1$	$\alpha = 5$
Error-free	ResNet	54.91%	65.73%	68.78%	CNN_1	55.69%	65.45%	70.47%
	MobileNet	56.96%	62.73%	69.23%	CNN_2	56.36%	64.99%	72.29%
	ShuffleNet	49.99%	58.85%	61.69%	CNN_3	57.17%	67.38%	72.22%
	Overall	53.95%	62.44%	66.57%	Overall	56.41%	65.94%	71.66%
Error-prone (mMIMO with ZF)	ResNet	51.77%	64.31%	66.73%	CNN_1	54.56%	63.77%	70.39%
	MobileNet	53.58%	61.16%	71.06%	CNN_2	52.15%	64.18%	71.04%
	ShuffleNet	46.02%	58.38%	59.47%	CNN_3	56.14%	66.18%	71.75%
	Overall	50.46%	61.23%	65.76%	Overall	54.28%	64.71%	71.06%

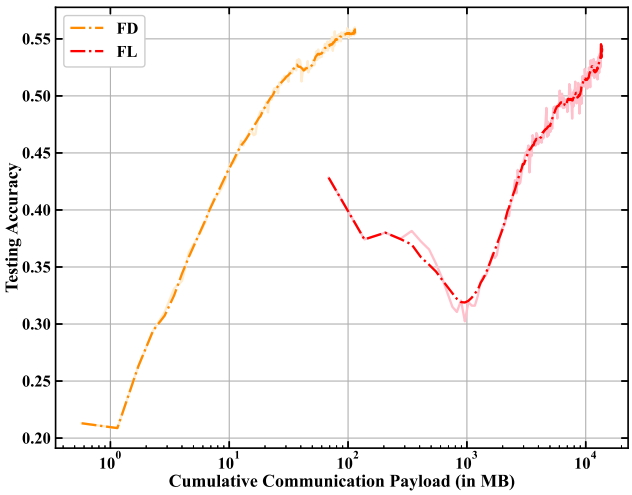

Fig. 10. Averaged testing accuracy versus the cumulative payload across communication rounds for FD and FL algorithms with fixed UL SNR −8 dB, DL SNR 3 dB and with MNIST/FMNIST datasets.

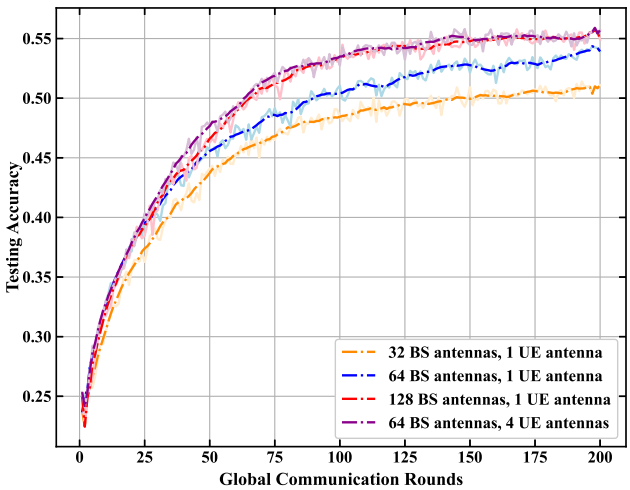

Fig. 11. Averaged testing accuracy across users for different numbers of BS/UE antennas at fixed UL-SNR −8 dB and DL-SNR 0 dB with CIFAR-10/STL-10 datasets.

local/private dataset and FMNIST is used for distillation. The smoothed moving average result with a window size of 3 is plotted in red. It can be seen that with the increase of the number of clients, the converged testing accuracy shows an increasing trend. However, such increment becomes smaller after $N = 15$, this is because in the massive MIMO communication system with the same number of antennas at BS and higher number of users, UL and DL SNRs decrease and consequently, the bit error rate (BER) will increase, leading to more communication errors in the received logits. Therefore, the gain from increasing the number of clients becomes negligible; that is, the system becomes saturated.

G. Communication Payload Comparison Between FD and FL

In Fig. 10, the testing accuracies of massive MIMO communication system-integrated FL [30] and FD algorithms are compared in terms of communication payload for fixed UL SNR −8 dB and DL SNR 3 dB with MNIST/FMNIST datasets. In the FL algorithm, each of 15 users has the same CNN model with 1,199,882 learnable weights and 16-bit quantization is used for communication. In the FD algorithm, with 8-bit quantization, the number of logit values is 2000 × 10 where 10 is the number of classes. For fair comparison, users in both frameworks are assumed to have the same CNN model architecture. Therefore, with much less communication payload in FD, the plot for FD converges much earlier than that of FL. The variations in the FL plot are at earlier rounds, which occur due to random initializations and local minimia of weights. At the convergence, it can be witnessed that the difference in the cumulative payload is approximately 14 GB, that is, FD overhead is around 1% of that of FL.

H. With Different Numbers of BS and UE Antennas

In Fig. 11, for CIFAR-10/STL-10 datasets, the numbers of BS and UE antennas are varied. Comparing yellow (32 BS antennas), blue (64 BS antennas), and red (128 BS antennas) curves, the performance improvement can be witnessed as the number of BS antennas is increased. As an illustration, testing accuracy experiences improvement from approximate 50% to 55% by augmenting the number of BS antennas from 32 to 128. Similarly, with the increase of UE antenna numbers, an obvious ascent in testing accuracy can be observed by comparing blue and purple curves, where UE antenna numbers are set to be 1 and 4, respectively.

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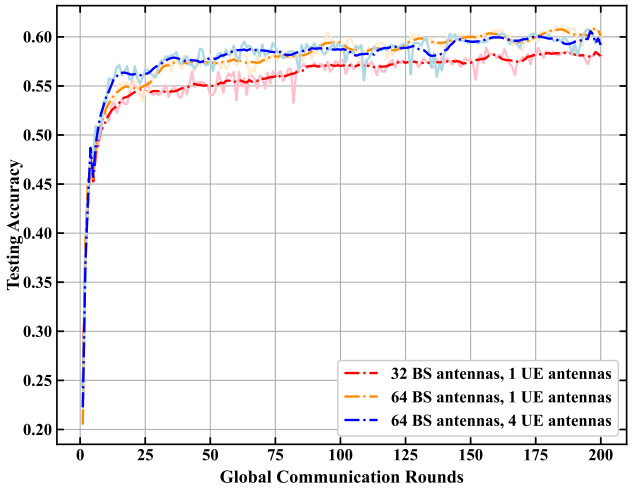

Fig. 12. Averaged testing accuracy across users for different numbers of BS/UE antennas at fixed UL-SNR -8 dB and DL-SNR 0 dB with ImageNette/STL-10 datasets.

I. Simulations on ImageNette Dataset

In Fig. 12, testing accuracies are plotted for ImageNette/STL-10 datasets with different numbers of BS/UE antennas. Due to the high-quality image in the ImageNette dataset, it can be witnessed that the model converges faster compared with CIFAR-10 or MNIST dataset scenarios.

VII. Conclusion

In this paper, we have integrated the federated distillation framework with multi-user massive MIMO system considering the imperfections of both UL and DL links. At the massive MIMO BS, ZF and MMSE schemes have been used to null the intra-cell interference. The convergence bound on the modeling error has been derived. By varying the number of local training steps in the proposed FedTSKD algorithm, we have improved the communication efficiency and accelerated the training process. Further, we have proposed a group-split scheme (FedTSKD-G) to let the model learn the imperfections introduced by the communication links intrinsically at the BS. The testing accuracy has been compared with the scenario with the perfect links. Comparison between FD and FL algorithms in terms of communication payload cost has also been included to demonstrate the significantly less overhead of FD, specifically, FD needs 1% of the FL's communication overhead.

Future work includes extending the proposed framework to the multi-cell scenario, conducting corresponding convergence analysis, and investigating more flexible aggregation strategies at the server side for the proposed FD framework.

Appendix A
Proof of Variance for the Total Effective Noise

Let $W = D_p \times C$. Then, the variance of the effective noise $\sum_{n=1}^{N} \lambda_n \bar{\omega}_{n,t}$ is obtained as

$$\begin{aligned}
W\sigma_\omega^2 &= \mathbb{E}\left\|\sum_{n=1}^{N} \lambda_n \bar{\omega}_{n,t}\right\|_2^2 = \mathbb{E}\left\|\sum_{n=1}^{N} \lambda_n \left(\bar{\omega}_{t,av} + \bar{\omega}_{n,t,DL}\right)\right\|_2^2 \\
&= \sum_{k,j=1}^{N} \lambda_k \lambda_j \, tr\mathbb{E}\left(\bar{\omega}_{t,av} + \bar{\omega}_{k,t,DL}\right)\left(\bar{\omega}_{t,av}^H + \bar{\omega}_{j,t,DL}^H\right) \\
&= \sum_{k,j=1}^{N} \lambda_k \lambda_j \, tr\mathbb{E}\left\{\bar{\omega}_{t,av}\bar{\omega}_{t,av}^H + \bar{\omega}_{k,t,DL}\bar{\omega}_{j,t,DL}^H\right\} \\
&= \left[\frac{\|\boldsymbol{\lambda}\|_2^2 \sigma_z^2}{2} \frac{\sigma^2 N_D}{P_{UL} N_{BS}} + \sigma_D^2\right] tr\mathbf{I}_W \\
&\quad + \sum_{n=1}^{N} \lambda_n^2 \, tr\mathbb{E}\left\{\bar{\omega}_{n,t,DL}\bar{\omega}_{n,t,DL}^H\right\} \\
&= \left[\frac{\|\boldsymbol{\lambda}\|_2^2 \sigma_z^2}{2} \frac{\sigma^2 N_D}{P_{UL} N_{BS}} + \sigma_D^2\right] W + \frac{\|\boldsymbol{\lambda}\|_2^2 \hat{\sigma}_z^2}{2} \frac{\sigma^2}{P_{DL}} W.
\end{aligned}$$

Appendix B
Proof of Lemma 1

The equations are given at the bottom of this page. In (a), the fact that $L_n(\mathbf{w}_n^*) \leq L_n(\mathbf{w}_{n,t})$ is used. The inequality in (b) is from the results of Assumptions 3 and 4.

$$\begin{aligned}
&\mathbb{E}\left\|\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\right\|_2^2 \leq \mathbb{E}\left\|\mathbf{w}_{n,t} - \eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right) - \mathbf{w}_n^*\right\|_2^2 \\
&= \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \mathbb{E}\left\|-\eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right)\right\|_2^2 - 2\Re\left(\mathbf{w}_{n,t} - \mathbf{w}_n^*\right)^T \eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right) \\
&= \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \mathbb{E}\left\|-\eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right) + \eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right\|_2^2 + \mathbb{E}\left\|-\eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right\|_2^2 \\
&\qquad -2\Re\mathbb{E}\left(\mathbf{w}_{n,t} - \mathbf{w}_n^*\right)^T \eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right) \\
&\qquad +2\Re\left(-\eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right)^T \underbrace{\mathbb{E}\left(-\eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right) + \eta_t \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right)}_{=0} \\
&= (1 - \eta_t \mu)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \eta_t^2 \mathbb{E}\left\|\nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right) - \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right)\right\|_2^2 + \eta_t^2 \mathbb{E}\left\|\nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right\|_2^2 \\
&\qquad +2\eta_t \mathbb{E}\left[L_n(\mathbf{w}_n^*) - L_n\left(\mathbf{w}_{n,t}\right)\right] \\
&\overset{(a)}{\leq} (1 - \eta_t \mu)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \eta_t^2 \mathbb{E}\left\|\nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right) - \nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_{n,t}\right)\right\|_2^2 + \eta_t^2 \mathbb{E}\left\|\nabla L_n\left(\mathbf{w}_{n,t}, \mathcal{D}_n\right)\right\|_2^2 \\
&\overset{(b)}{\leq} (1 - \eta_t \mu)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \eta_t^2 \left(\sigma_n^2 + G^2\right)
\end{aligned}$$

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APPENDIX C
PROOF OF COROLLARY 1

We conduct our proof via induction as follows. Let $C_n = (\sigma_n^2 + G^2), \nu_n \geq \frac{\beta_0^2 C_n}{\beta_0\mu - 1}, \eta_t = \frac{\beta_0}{t+\beta_1}, \beta_0 > \frac{1}{\mu}, \beta_1 > 0$. By utilizing the results from Lemma 1, for $t \geq 0$, we can get the bound for $\mathbb{E}\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\|_2^2$ as follows

$$\mathbb{E}\left\|\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\right\|_2^2 \leq (1-\eta_t\mu)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \eta_t^2 C_n \tag{29}$$

$$\leq \frac{\nu_n}{t+\beta_1} - \frac{\beta_0\mu\nu_n}{(t+\beta_1)^2} + \frac{\beta_0^2 C_n}{(t+\beta_1)^2} \tag{30}$$

$$= \frac{\nu_n(t+\beta_1) - \nu_n}{(t+\beta_1)^2} + \frac{\nu_n - \beta_0\mu\nu_n + \beta_0^2 C_n}{(t+\beta_1)^2} \tag{31}$$

$$\leq \frac{\nu_n}{t+1+\beta_1} + 0. \tag{32}$$

APPENDIX D
PROOF OF LEMMA 2

The equations are given at the bottom of this page. In (a), μ_q strong convexity from Assumption 2 is used. In (b), results from Assumptions 3 and 4 are utilized. The inequality in (c) comes from the fact that $Q_n(\mathbf{w}_n^*) \leq Q_n(\mathbf{w}_n)$.

$$\begin{aligned}
&\mathbb{E}\left\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\right\|_2^2 = \mathbb{E}\left\|\mathbf{w}_{n,t} - \sum_{i=0}^{S-1}\eta_{t+S+i}\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) - \mathbf{w}_n^*\right\|_2^2 \\
&= \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\left(\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) \pm \nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right)\right\|_2^2 \\
&\quad -2\Re\mathbb{E}\sum_{i=0}^{S-1}\eta_{t+S+i}\left(\mathbf{w}_{n,t} - \mathbf{w}_n^*\right)^T\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) \\
&= \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\left(\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) - \nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right)\right\|_2^2 \\
&\quad +\mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right\|_2^2 \\
&\quad -2\Re\sum_{i=0}^{S-1}\eta_{t+S+i}\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)^T\left[\sum_{i=0}^{S-1}\eta_{t+S+i}\underbrace{\mathbb{E}\left(\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) - \nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right)}_{=0\text{, based on SGD}}\right] \\
&\quad -2\Re\sum_{i=0}^{S-1}\eta_{t+S+i}\mathbb{E}\left(\mathbf{w}_{n,t} - \mathbf{w}_n^*\right)^T\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) \\
&= \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\left(\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) - \nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right)\right\|_2^2 \\
&\quad +\mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right\|_2^2 - 2\Re\sum_{i=0}^{S-1}\eta_{t+S+i}\mathbb{E}\left(\mathbf{w}_{n,t} - \mathbf{w}_n^*\right)^T\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_{n,t+S+i}\right) \\
&\overset{(a)}{\leq} \mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \left(\sum_{i=0}^{S-1}\eta_{t+S+i}^2\right)S\sigma_{qn}^2 + \mathbb{E}\left\|\sum_{i=0}^{S-1}\eta_{t+S+i}\nabla Q_n\left(\mathbf{w}_{n,t+S+i}, \hat{\mathcal{Z}}_n\right)\right\|_2^2 \\
&\quad +2\Re\sum_{i=0}^{S-1}\eta_{t+S+i}\left[\mathbb{E}\left\{Q_n\left(\mathbf{w}_n^*, \hat{\mathcal{Z}}_{n,t+S+i}\right) - Q_n\left(\mathbf{w}_{n,t}, \hat{\mathcal{Z}}_{n,t+S+i}\right)\right\} - \frac{\mu_q}{2}\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2\right], \\
&\overset{(b)}{\leq} \left(1 - \mu_q\sum_{i=0}^{S-1}\eta_{t+S+i}\right)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \left(\sum_{i=0}^{S-1}\eta_{t+S+i}^2\right)\left(S\sigma_{qn}^2 + SG_q^2\right) \\
&\quad +2\Re\sum_{i=0}^{S-1}\eta_{t+S+i}\underbrace{\left[\mathbb{E}\left\{Q_n\left(\mathbf{w}_n^*, \hat{\mathcal{Z}}_{n,t+S+i}\right) - Q_n\left(\mathbf{w}_{n,t}, \hat{\mathcal{Z}}_{n,t+S+i}\right)\right\}\right]}_{Q_n(\mathbf{w}_n^*) - Q_n(\mathbf{w}) \leq 0} \\
&\overset{(c)}{\leq} \left(1 - \mu_q\sum_{i=0}^{S-1}\eta_{t+S+i}\right)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 + \left(\sum_{i=0}^{S-1}\eta_{t+S+i}^2\right)S\left(\sigma_{qn}^2 + G_q^2\right)
\end{aligned}$$

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Appendix E
Proof of Corollary 2

Similar to Corollary 1, we conduct our proof via induction as follows. Let $C_{qn} = (\sigma_{qn}^2 + G_q^2), \nu_n \geq \frac{S^2\beta_0^2 C_{qn}}{\beta_0\mu_q - 2S}, \eta_t = \frac{\beta_0}{t+\beta_1}$, where $\beta_0 > \frac{2S}{\mu_q}, \beta_1 \geq 4$. By utilizing the results from Lemma 2, for $t \geq 0$, we write

$$\begin{aligned}
\mathbb{E}\left\|\mathbf{w}_{n,t+2S} - \mathbf{w}_n^*\right\|_2^2 &\leq \left(1 - \mu_q \sum_{i=0}^{S-1} \eta_{t+S+i}\right)\mathbb{E}\left\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\right\|_2^2 \\
&\quad + \left(\sum_{i=0}^{S-1} \eta_{t+S+i}^2\right) SC_{qn} \\
&\leq \left(1 - \mu_q \sum_{i=0}^{S-1} \eta_{t+S+i}\right)\frac{\nu_n}{t+\beta_1} + \left(\sum_{i=0}^{S-1} \eta_{t+S+i}^2\right) SC_{qn} \\
&\leq \frac{\nu_n}{t+\beta_1} - \frac{\nu_n}{t+\beta_1}\mu_q \sum_{i=0}^{S-1} \eta_{t+S+i} + \left(\sum_{i=0}^{S-1} \eta_{t+S+i}^2\right) SC_{qn} \\
&\overset{(a)}{\leq} \frac{\nu_n}{t+\beta_1} \times \frac{t+\beta_1}{t+\beta_1} - \frac{\nu_n}{t+\beta_1}\frac{\beta_0}{t+\beta_1}\mu_q + \frac{S\beta_0^2}{(t+\beta_1)^2} SC_{qn} \\
&= \frac{\nu_n(t+\beta_1) - 2S\nu_n}{(t+\beta_1)^2} + \frac{2S\nu_n}{(t+\beta_1)^2} - \frac{\nu_n}{t+\beta_1}\frac{\beta_0}{t+\beta_1}\mu_q \\
&\quad + \frac{S\beta_0^2}{(t+\beta_1)^2} SC_{qn} \\
&\leq \frac{\nu_n}{t+2S+\beta_1} + \frac{2S\nu_n - \nu_n\beta_0\mu_q + S^2\beta_0^2 C_{qn}}{(t+\beta_1)^2} \\
&\leq \frac{\nu_n}{t+2S+\beta_1} + 0,
\end{aligned}$$

where in (a), the following results are used

$$\begin{aligned}
\sum_{i=0}^{S-1} \eta_{t+S+i} &= \sum_{i=0}^{S-1} \frac{\beta_0}{t+S+i+\beta_1} \overset{(b)}{>} \frac{\beta_0}{t+\beta_1} \\
\sum_{i=0}^{S-1} \eta_{t+S+i}^2 &= \sum_{i=0}^{S-1} \frac{\beta_0^2}{(t+S+i+\beta_1)^2} < \frac{S\beta_0^2}{(t+\beta_1)^2},
\end{aligned}$$

in inequality (b), the condition that $\beta_1 \geq 4$ is used.

Appendix F
Proof of Theorem 1

With the ψ-smoothness in Assumption 1, we write

$$\begin{aligned}
&\mathbb{E}\left\{L\left(\mathbf{w}_{n,t+1}\right)\right\} - L_n^* \\
&\leq \mathbb{E}\left\{\left(\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\right)^T \nabla L(\mathbf{w}_n^*)\right\} + \frac{\psi}{2}\mathbb{E}\left\|\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\right\|_2^2 \\
&\overset{(a)}{=} \frac{\psi}{2}\mathbb{E}\left\|\mathbf{w}_{n,t+1} - \mathbf{w}_n^*\right\|_2^2 \\
&\overset{(b)}{\leq} \frac{\psi}{2} \cdot \frac{\nu_n}{t+1+\beta_1},
\end{aligned}$$

where in (a), zero mean result $\mathbb{E}\{\mathbf{w}_{n,t} - \mathbf{w}_n^*\} = \mathbf{0}$ is used. In (b), the inequality, $\mathbb{E}\|\mathbf{w}_{n,t} - \mathbf{w}_n^*\|_2^2 \leq \frac{\nu_n}{t+\beta_1}$ is from Corollary 1 and Corollary 2, where $\nu_n \geq \max(\frac{\beta_0^2 C_n}{\beta_0\mu - 1}, \frac{S^2\beta_0^2 C_{qn}}{\beta_0\mu_q - 2S})$, $C_{qn} = (\sigma_{qn}^2 + G_q^2), C_n = (\sigma_n^2 + G^2), \eta_t = \frac{\beta_0}{t+\beta_1}, \beta_0 > \max(\frac{1}{\mu}, \frac{2S}{\mu_q}), \beta_1 \geq 4$.

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